ROCHAK GHIMIRE

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EDUCATION

Morgan State University Baltimore, MD

B.S. Computer Science, GPA: ***3.947****/4.00* *Expected May 2028*

- **Dean's List:** Every semester, Fall 2024–Spring 2026 | Clara Adams Honors College Scholar
- **Awards:** 1st Place, CEAMLS Human-AI Collaboration Challenge – ReadAloud (offline K-12 literacy assessment tool)
- **Relevant Coursework:** Data Structures, Algorithms Analysis, Artificial Intelligence, Database Management, Networking, Cyber Security

EXPERIENCE

Human-AI eXperience (HAX) Lab Sep. 2026 – Present

Undergraduate Researcher, Morgan State University

- Translate user-research findings into system requirements and prototype **interactive AI systems** using **human-centered design** and **HCI** principles.
- Contribute to literature reviews, **system design**, and **evaluation** for trustworthy, accessible human-AI interaction.

Center for Equitable AI and Machine Learning Systems (CEAMLS) May 2026 – Jul. 2026

Undergraduate ML Research Intern, Morgan State University

- Designed and **trained** a multimodal concept-drift detection model for early cardiovascular disease prediction, fusing a **1D CNN**, **2D CNN**, and **Transformer** over time, time-frequency, and frequency domains of ECG/PPG time-series (PyTorch), achieving **87.86% drift-classification accuracy** and **0.8446** energy-drift correlation.
- Built a cross-modal Complementary Feature Domain (CFD) **feature-engineering pipeline** (pyCFD, NumPy, Pandas, scikit-learn), processing **30,000+** physiological signals from PTB-XL and MIMIC over **four** benchmarks.
- Evaluated generalization on held-out MIMIC waveforms, outperforming ADWIN, Page-Hinkley, MMD, and Dropout Uncertainty baselines by up to **40 percentage points** in drift-classification accuracy.

PROJECTS

MyVoice: AI Self-Advocacy Coach | *React, Node/Express, Claude API, ElevenLabs* | GitHub Jul. 2026

- Architected a full-stack generative-AI coaching platform for K-12 students with IEPs/504 plans, orchestrating **four Claude model roles** (Haiku, Sonnet) across a **4-stage workflow**, with ElevenLabs voice synthesis for roleplay.
- Built a **tool-use-constrained classifier** mapping free-text input to **18 accommodation categories**, with **structured outputs** enforced at the API level.
- Hardened the Express backend with **per-IP rate limiting**, CORS allowlisting, and **40-turn context truncation**, plus a zero-key mock mode for cost-free end-to-end testing.

Redis Clone: In-Memory Key-Value Server | *Python, asyncio, RESP Protocol* | GitHub Aug. 2026

- Built a **Redis-compatible async backend server** from raw TCP sockets, hand-writing the **RESP2 protocol**; the real redis-py client works with it unmodified.
- Implemented core commands plus **TTL/EXPIRE** (passive and active expiration), with **zero lost updates** in stress tests with **50+ concurrent clients**.
- Added **append-only persistence** and **primary-replica replication**, verified by mid-write crash recovery in a pytest suite of **40+ tests**.
- Benchmarked **101,010 req/s** reads at sub-ms p50 and ~**23,000 req/s** durable writes, and traced the write bottleneck to per-command `fsync()`.

MoodLens: Wellness Tracking Application | *React, Firebase, Groq API* | GitHub Feb. 2026

- Built a wellness tracker (**React 19, Firebase, Firestore**) with mood, sleep, and stress logging, a **30-day mood heatmap**, and **UID-scoped security rules**.
- Integrated **Llama 3.1 via the Groq API** to generate weekly and monthly wellness reports, with **per-user response caching** to avoid redundant API calls.
- Added code splitting (**8 lazy-loaded chunks**), **IndexedDB offline persistence**, and a zero-signup demo mode.

TECHNICAL SKILLS

Languages: Python, JavaScript, Java, C++, SQL
Frameworks & Libraries: PyTorch, scikit-learn, NumPy, Pandas, React, Node.js/Express, Firebase, asyncio, pytest
Tools & Practices: Git, GitHub Actions (CI/CD), Docker, Vite, REST APIs, System Design
AI/ML: Deep Learning (CNNs, Transformers), Model Training & Evaluation, Multimodal Learning, Time-Series Signal Processing, NLP, LLM Orchestration, Structured Outputs